

Motor test patterns

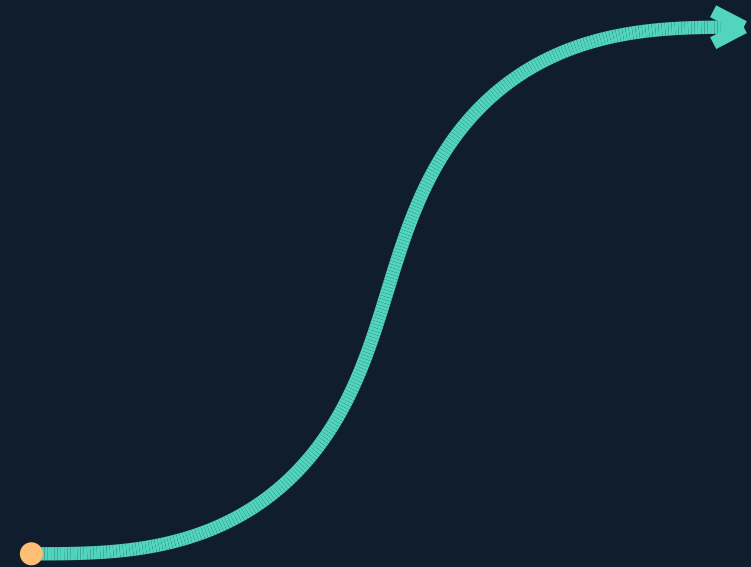
Usage, motion meaning and result interpretation

**One node. Five patterns.
A repeatable way to inspect
motor and platform behavior.**

Straight / Rotate / Circle / Square / S

Commands: aic mc tp <pattern>

Editable guide + printable reference



Ideal S reference - not test data

Choose the question before the pattern

All commands execute inside the existing `motor_control_node`.

straight	Forward + reverse	Wheel balance; reverse response
rotate	Left + right in place	Turn symmetry; angle repeatability
circle	Constant-radius lap	Wheel-speed ratio; radial drift
square	4 lines + 4 stopped turns	Distance/angle error accumulation
s	Smooth left/right bend	Transition response; path correction

Default repetitions: 3 pairs for straight/rotate; 3 complete shapes for circle/square/S.

Start a test from standstill

Build/install the updated node and deploy the updated AICamera helper on the robot.

```
ros2 run motor_control_g4dual motor_control_node \  
  --ros-args --params-file /path/to/motor_control.yaml \  
  -p enable_can:=true
```

One shell command; replace the parameter-file path.

- 1 Stop periodic cmd_vel publishers.
- 2 aic mc en -> aic mc e diag
- 3 Wait for fresh feedback + standstill.
- 4 aic mc tp straight

Monitor in another terminal

aic mc tp status

Required before starting

CAN on; motors enabled; no fault latch.

Previous command must expire

Default cmd_vel timeout: 500 ms.

RPM resolution: 1.0 RPM/unit.

Confirmed for this project.

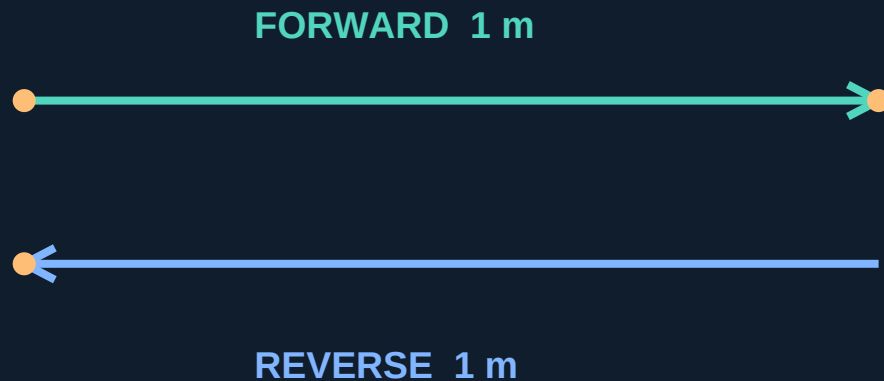
YAML, launch and CLI agree.

Other projects are unchanged.

Straight: compare forward and reverse

One repetition = forward 1 m, stop, reverse 1 m, stop.

```
aic mc tp straight
```



Same physical line; paths offset here for clarity.
Amber dots indicate a standstill check.

What it tells you

Left/right wheel balance and reverse response.

Default motion

0.1 m/s; 3 forward/reverse pairs.

How it finishes

Distance projected onto each segment heading.

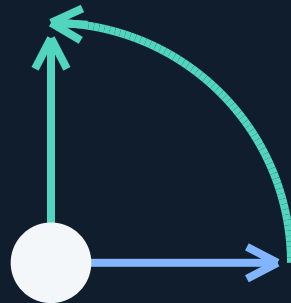
Interpretation

Return error is measured, not corrected.

Rotate: isolate turning behavior

One repetition = left 90 degrees, stop, right 90 degrees, stop.

```
aic mc tp rotate
```



LEFT 90 deg / RIGHT 90 deg

What it tells you

Turning symmetry and repeatable angles.

Default motion

0.2 rad/s; 3 left/right pairs.

Configurable angle

angle_deg supports 90, 180 or 360 degrees.

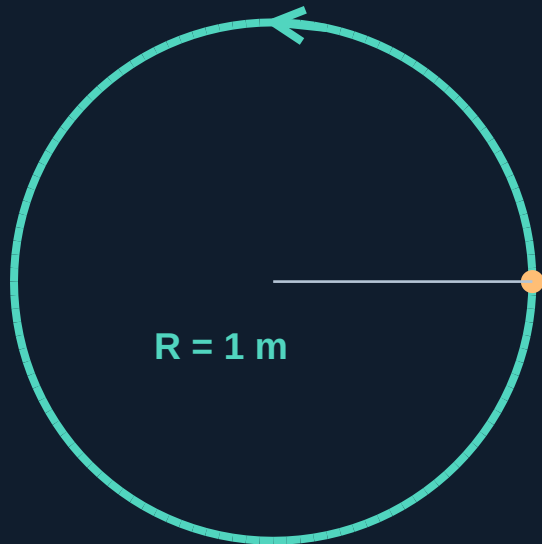
How it finishes

Continuous yaw handles the +/-180 boundary.

Circle: compare inner and outer wheels

One repetition = one full lap, then standstill. Default: counterclockwise.

```
aic mc tp circle
```



Robot-centre radius; wheel paths differ.

What it tells you

Wheel-speed ratio and radius consistency.

Default motion

Radius 1 m; 3 laps; about 64 s per lap.

Motion relationship

$v = \text{radius} \times \text{angular speed}$; limits apply.

How it finishes

Accumulated yaw reaches 360 deg, then stop.

Square: expose accumulated error

One repetition = four forward sides and four stationary 90-degree turns.

```
aic mc tp square
```



Stop after every line and turn; schematic only.

What it tells you

Distance, turn and stop errors accumulate.

Default motion

1 m sides; counterclockwise; 3 squares.

Fixed corners

Always 90 deg; angle_deg only affects rotate.

How it finishes

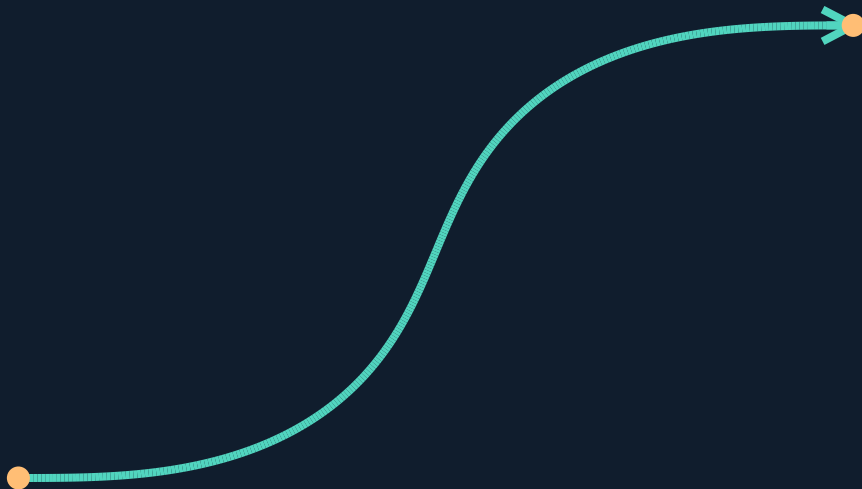
Includes the last turn toward the start heading.

Smooth S: test transition and correction

One repetition = a continuous left/right bend; stop only at the end.

aic mc tp s

NO STOP AT THE MIDDLE



Ideal reference; end is displaced from start.

What it tells you

Smooth turning response and tracking error.

Default geometry

4 m travel length; 1 m minimum nominal radius.

Path correction

Heading + lateral gains default to 1.0.

For motor-only response

Set both S gains to 0, then restart the node.

Change geometry, limits and direction

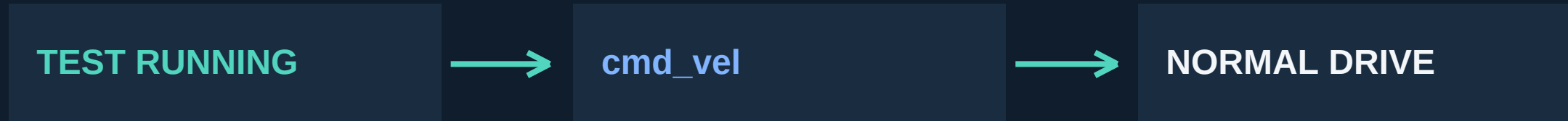
Edit `motion_test.*` in `motor_control.yaml`, then restart the node.

Parameter suffix	Default	Meaning
<code>distance_m</code> / <code>angle_deg</code>	1.0 / 90.0	Straight segment / rotate angle
<code>circle_radius_m</code> / <code>square_side_m</code>	1.0 / 1.0	Circle radius / square side
<code>s_length_m</code> / <code>s_radius_m</code>	4.0 / 1.0	S travel length / minimum nominal radius
<code>clockwise</code>	false	Circle/square clockwise; S right-first if true
<code>linear_speed_mps</code> / <code>angular_speed_radps</code>	0.1 / 0.2	Body velocity limits
<code>linear_accel_mps2</code> / <code>angular_accel_radps2</code>	0.1 / 0.2	Body acceleration limits
<code>s_heading_gain</code> / <code>s_lateral_gain</code>	1.0 / 1.0	S correction: 1/s and 1/m ²
<code>repetitions</code>	3	Pairs or complete shapes, depending on pattern

The feature defaults to enabled. It does not start tests or enable motors automatically.

Normal commands can take over

Tests and normal driving share the same node and motor enable state.



Valid cmd_vel, including zero

Ends the test and takes control with that command.

Normal completion

Zero speed; motors stay enabled for the next command.

Cancel, fault or emergency stop

Motion is gated; explicitly re-enable after resolving it.

Pause periodic `cmd_vel` publishers while testing, or their next message will interrupt the test.

Read the result, not only the status

aic mc tp status shows state, segment, outcome reason and the CSV path.

ILLUSTRATIVE STOPPED RESULT

Target	1.000 m
Measured	1.025 m
Difference	+0.025 m

+25 mm overshoot

```
last_stopped_segment_progress  
- last_stopped_segment_target
```

Check units and segment type

line / s_curve: m turn / arc: rad

Compare targets and feedback

Normalized wheel RPM; account for feedback delay.

Inspect path_error_m

Lateral / radial / turn translation / S cross-track.

CSV directory on the robot

/tmp/motor_control_tests (configurable)

completed = sequence finished. Wheel odometry is not independent proof of ground-track accuracy.

If a test will not start or stops early

A published command is a request. Check status for acceptance or the reason for interruption.

Rejected before start	Check enabled motors, CAN, fresh position + speed feedback.
Recent cmd_vel / moving wheels	Stop command publishers; wait for timeout and standstill.
Aborted: external cmd_vel	Normal command took control; the test will not resume.
No progress / stale feedback	Inspect obstruction, wheel direction, encoders and CAN.
Segment timeout	60 s per movement/wait; circle motion has a 120 s default.
CSV cannot be written	Use a writable local log directory; check disk space.

Feedback timeout: 500 ms. Stall timeout: 5 s. Standstill: both wheels ≤ 0.5 RPM for 0.5 s.
Cancel requests the existing ramp stop; an aborted state does not confirm physical standstill.

Commands to keep beside the robot

Use one pattern at a time. Begin with short, low-speed runs and inspect the record.

`aic mc tp straight`

Forward / reverse

`aic mc tp rotate`

Left / right rotation

`aic mc tp circle`

Full circle

`aic mc tp square`

Stopped-corner square

`aic mc tp s`

Smooth S

`aic mc tp status`

Monitor outcome + CSV

`aic mc tp cancel`

Cancel and gate motion

`aic mc tp help`

Show command help

Observe -> compare -> tune -> repeat. Validate physical accuracy using external measurements.